Exponential functions

1

a 5 b 80 c $\frac{5}{4}$ d 320

2

a 1 b $\frac{9}{16}$

3 $50\,313.28$

4

a 125 b 1 c $\frac{1}{5}$ d -2

5

a $(5,-32)$ b $\left(-3,-\frac{1}{8}\right)$ c $\left(-1,-\frac{1}{2}\right)$ d $(4,-16)$

6 $m=8$

7

a $n=8$ b $k=4$ c $m=-2$

Characteristics of exponential functions

8 No, both functions are always greater than zero.

9 None, for all values of x it will have a positive value.

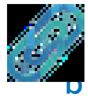
10

a $(0,1)$ b $(0,1)$ c $(0,-1)$ d $(0,-1)$

11

a 1 b 1

12



- **a** $\dfrac{1}{81}$
- **b** 1
- **c** 81
- **d** It gets larger, approaching infinity.
- **e** It gets smaller, approaching zero.

13

- **a** Positive
- **b** 0
- **c** $y = 0$ is a horizontal asymptote of the curve.

14 There is no maximum function value. As x decreases, y approaches 0 but never quite gets there.

15

- **a** $y = 4^x$
- **b** y is increasing at an increasing rate.

16 $y = 5^x$

17 $y = 5^{-x}$

18

a

x	-5	-4	-3	-2	-1	0	1	2	3	4	5	10
y	$\frac{1}{243}$	$\frac{1}{81}$	$\frac{1}{27}$	$\frac{1}{9}$	$\frac{1}{3}$	1	3	9	27	81	243	$59\,049$

- **b** As x increases, the function increases at a faster and faster rate.
- **c** All real x
- **d** $y > 0$

19

a

x	-5	-4	-3	-2	-1	0	1	2	3	4	5	10
y	32	16	8	4	2	1	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{16}$	$\frac{1}{32}$	$\frac{1}{1024}$

b As x increases, the function decreases at a slower and slower rate.

c 1

d All real x

e $y > 0$

20

a

x	-5	-4	-3	-2	-1	0	1	2	3	4	5	10
y	$-\frac{1}{32}$	$-\frac{1}{16}$	$-\frac{1}{8}$	$-\frac{1}{4}$	$-\frac{1}{2}$	-1	-2	-4	-8	-16	-32	-1024

b No. For $a > 0$ the expression a^x is greater than zero for all real x. Now, $-\left(a^x\right)$ will be less than or equal to zero.

c Decreasing

d As x increases, the function decreases at a faster and faster rate.

e All real x

f $y < 0$

21

a $(0, 1)$

b No

c All real x

d $y > 0$

e $y = 128$

f $x = 8$

22

a Example answers:

- None of the curves cross the x-axis.
- They all have the same y-intercept.
- As x gets large, the function values approach 0.

b 1

c As x gets large, the function values approach 0.

23

a $(0,1)$

b As x gets large $y=3^x$ becomes large while $y=3^{-x}$ approaches 0.

24

a No b Yes c Yes d No

Graphs of exponential functions

25

a Yes b No c No d No

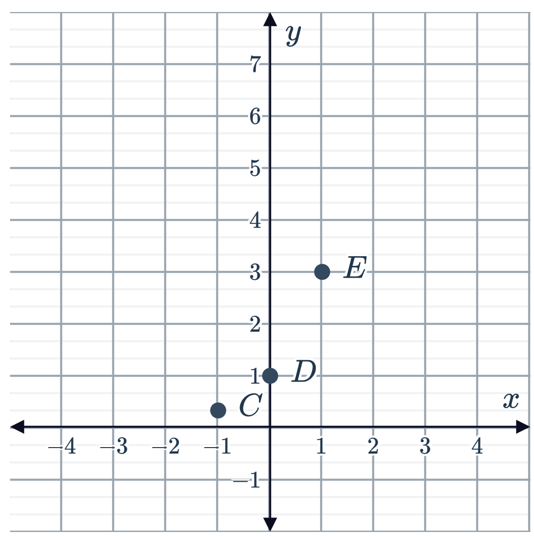
e No f No g No h Yes

26

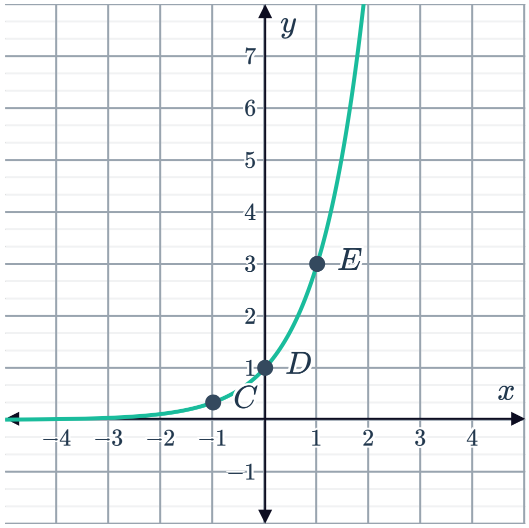
a

Point	Coordinates
A	$\left(-3, \frac{1}{27}\right)$
B	$\left(-2, \frac{1}{9}\right)$
C	$\left(-1, \frac{1}{3}\right)$
D	$(0,1)$
E	$(1,3)$
F	$(2,9)$
G	$(3,27)$

b



c

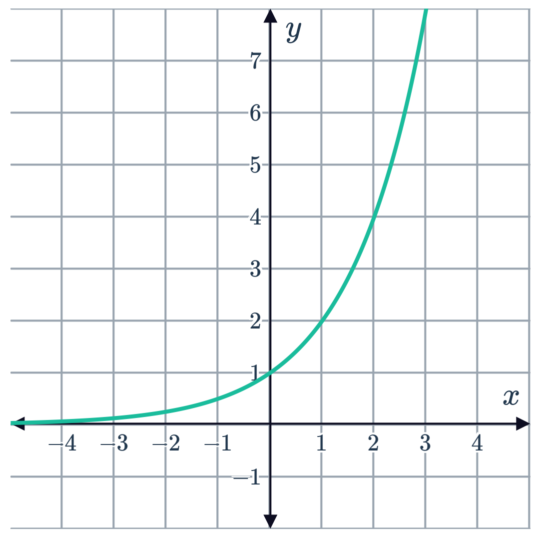


27

a

x	-3	-2	-1	0	1	2	3
y	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{2}$	1	2	4	8

b



c Increases

d Decreases

e The curve never crosses the x-axis.

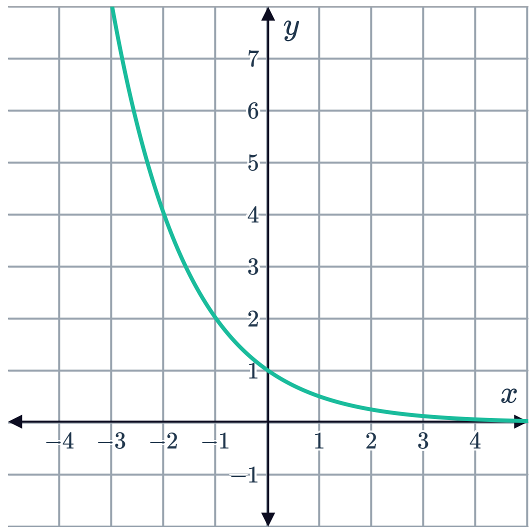
f 1

28

a

x	-3	-2	-1	0	1	2	3
y	8	4	2	1	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$

b



c Decreases

d Increases

e The curve never crosses the x-axis.

f 1

29

a

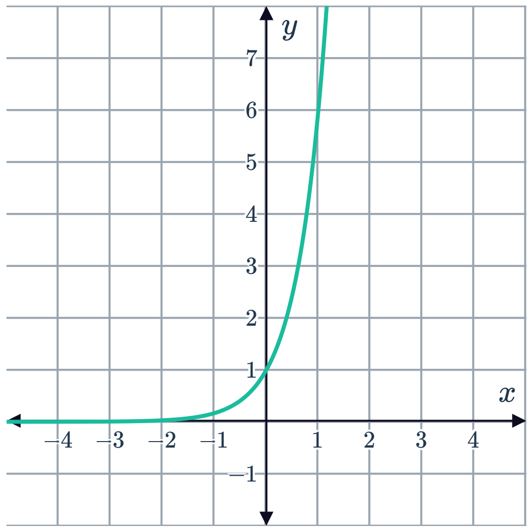
i $(0, 1)$

ii

x	-3	-2	-1	0	1	2	3
y	$\frac{1}{216}$	$\frac{1}{36}$	$\frac{1}{6}$	1	6	36	216

iii $y = 0$

iv



b

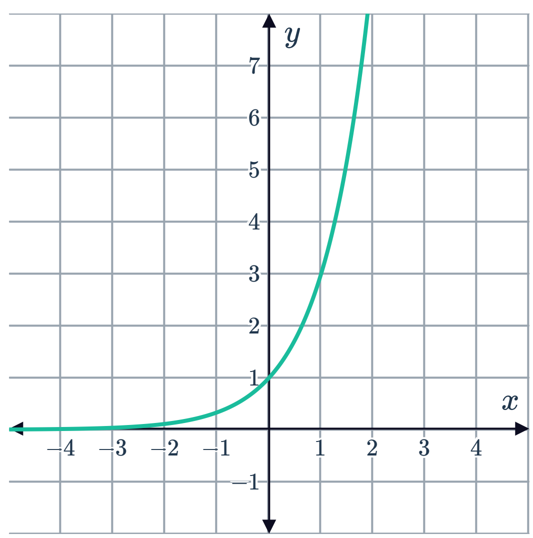
i $(0, 1)$

ii

x	-3	-2	-1	0	1	2	3
y	$\frac{1}{27}$	$\frac{1}{9}$	$\frac{1}{3}$	1	3	9	27

iii $y = 0$

iv



c

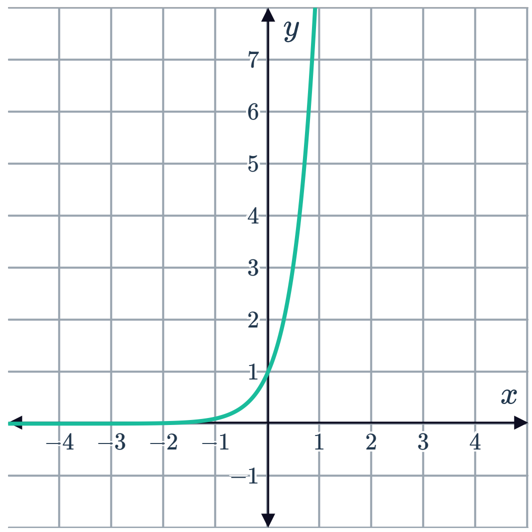
i $(0, 1)$

ii

x	-3	-2	-1	0	1	2	3
y	$\frac{1}{1000}$	$\frac{1}{100}$	$\frac{1}{10}$	1	10	100	1000

iii $y = 0$

iv



d

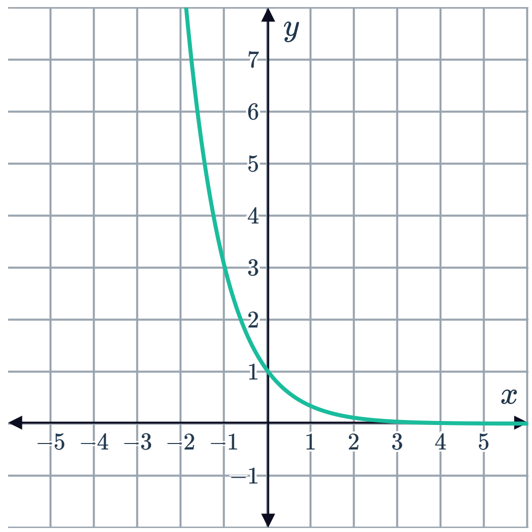
i $(0, 1)$

ii

x	-3	-2	-1	0	1	2	3
y	27	9	3	1	$\frac{1}{3}$	$\frac{1}{9}$	$\frac{1}{27}$

iii $y = 0$

iv



30

a No. For $a > 0$ the expression a^x is greater than zero for all real x.

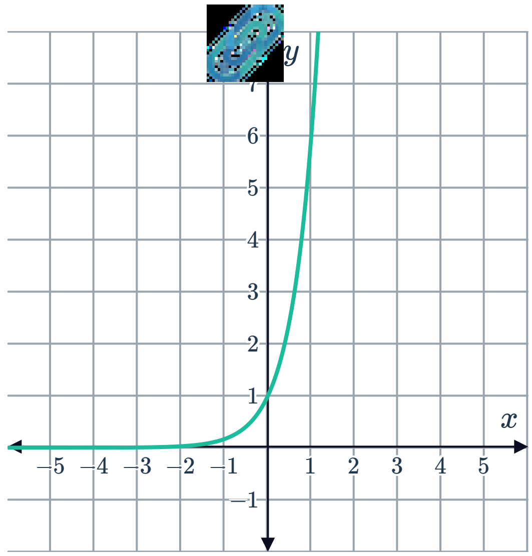
b ∞

c 0

d 1

e 0

f



31

a No. For $a > 0$ the expression a^x is greater than zero for all real x.

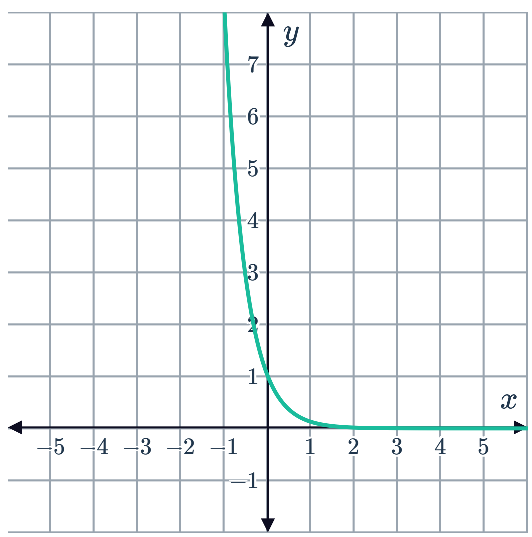
b 0

c ∞

d 1

e 0

f

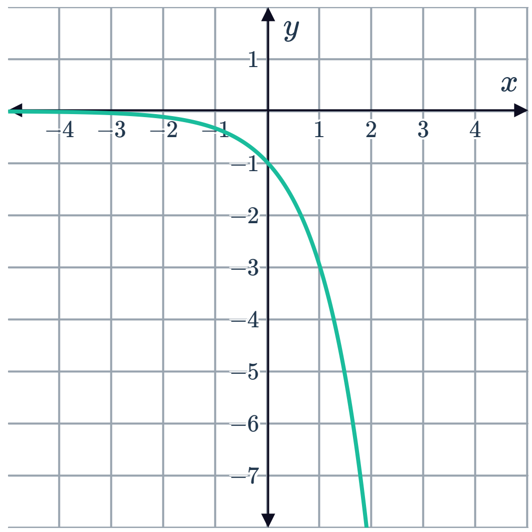


32

a $(0,-1)$

b $y=0$

c



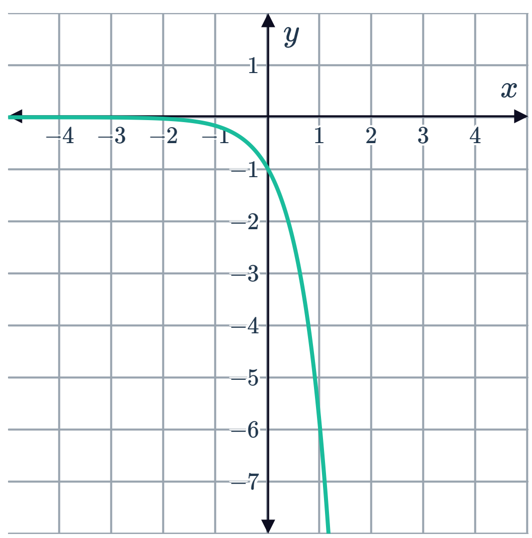
d Decreasing function

33

a $(0,-1)$

b $y=0$

c



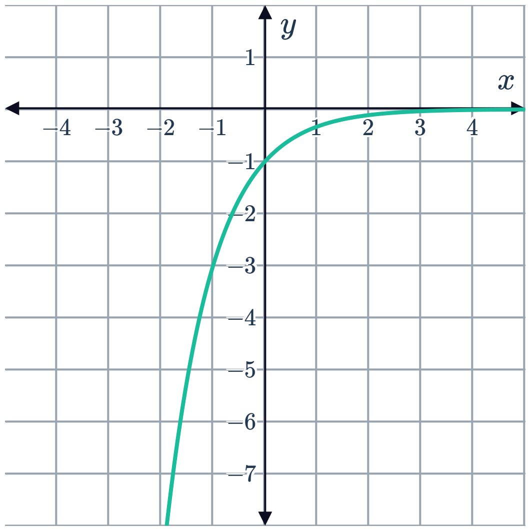
d Decreasing function

34

a $(0, -1)$

b $y = 0$

c



d Increasing function

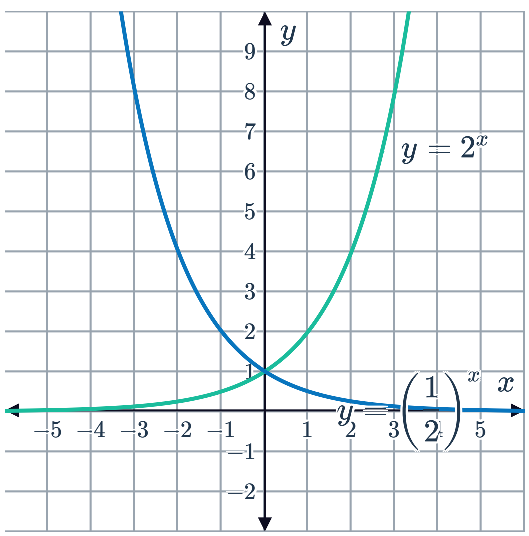
35

a

i Yes **ii** Yes **iii** No **iv** No

b Reflection across the y-axis.

c

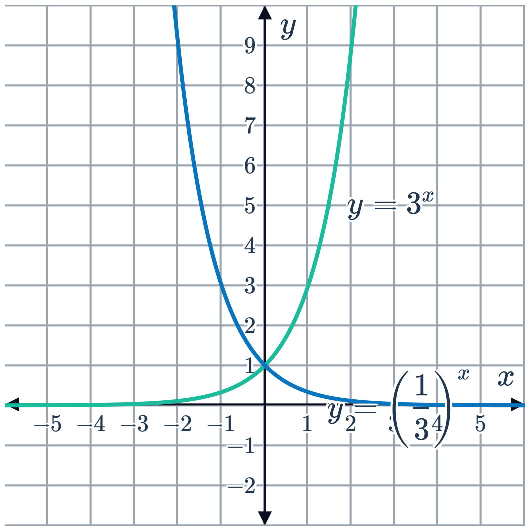


36

a $y = 3^{-x}$

b Reflection about the y-axis.

c



37

a $y = 0$

b $y = 0$

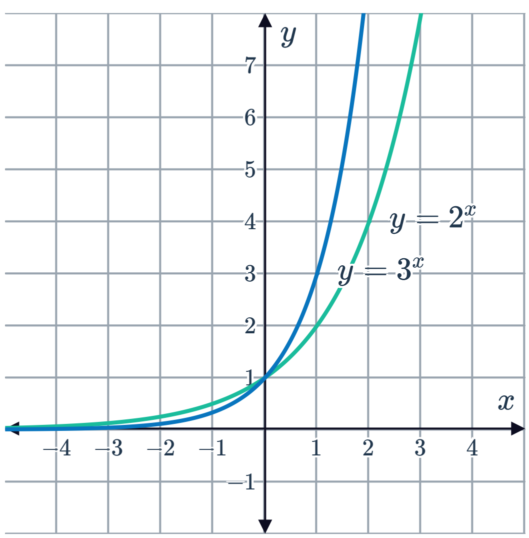
c

x	-3	-2	-1	0	1	2	3
y	$\frac{1}{27}$	$\frac{1}{9}$	$\frac{1}{3}$	1	3	9	27

d

x	-3	-2	-1	0	1	2	3
y	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{2}$	1	2	4	8

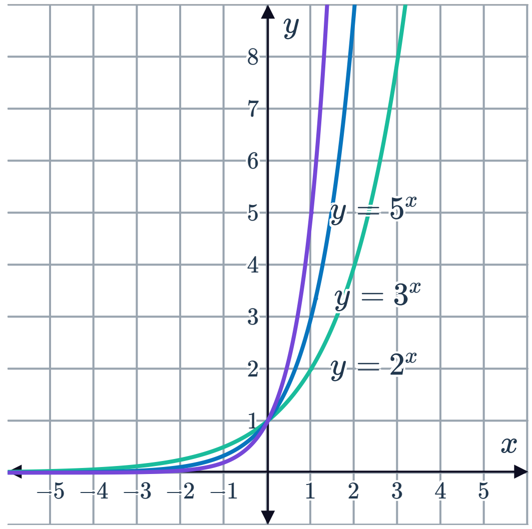
e



f $(0, 1)$

38

a



b Shared y-intercept of $(0, 1)$

c As x gets larger, the function increases at an increasing rate and the values approach ∞.

39

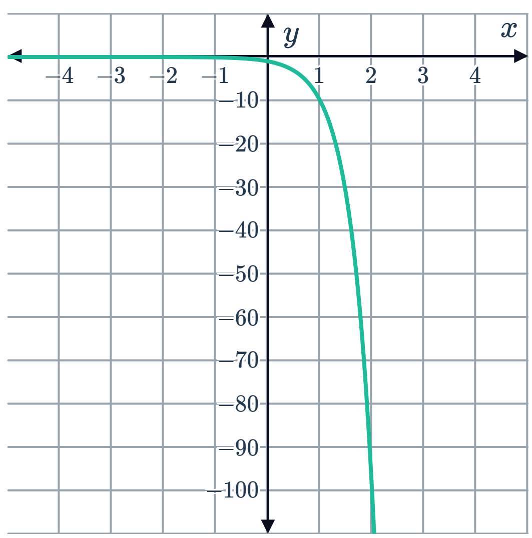
a Graph R

b For $x < 0$, the graph of $y = 6^x$ is below the graph of $y = 4^x$. This is because negative values of x result in fractional function values, and for any negative value of x, 6^x will result in a smaller value than 4^x.

40

a Yes. For any value of x, 10^x is always positive. Therefore, -10^x is always negative.

b



c No solution